MATHEMATICS OF CHANCE

PYQ • 2021 - 2026

1. 2026 **3 MARKS**

(A) A person is asked to say a three digit number.

(i) What is the probability that all the three digits of this number are the same ? **(1)**

(ii) What is the probability that it is a multiple of 5, which is less than 200 ? **(2)**

OR

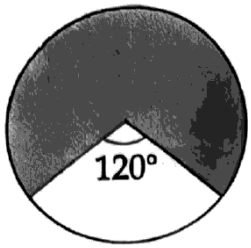
(B) A box contains four slips numbered 1, 3, 5, 7 and another box contains three slips numbered 2, 4, 6. If one slip is drawn from each box.

(i) What is the probability that the product of the numbers being a prime ? **(2)**

(ii) What is the probability that the product of the numbers is a multiple of 7 ? **(1)**

2. 2026 **2 MARKS**

In the figure, a sector of the circle with central angle 120° is given. If a dot is put in the circle, without looking, what is the probability that the dot is in the shaded region ?



3. 2025 **4 MARKS**

A box contains red and blue beads. 16 out of them are red. If the probability of getting a red bead is 2/3,

(a) What is the total number of beads ?

(b) What is the probability of getting a blue bead ?

(c) If 4 red beads are removed from the box, what is the probability of getting a red bead ?

4. 2025 **2 MARKS**

Numbers from 1 to 20 are written on paper slips and put in a box. If a slip is to be taken from it without looking,

(a) What is the probability that it is a prime number ?

(b) What is the probability that it is a perfect square ?

5. 2024 **4 MARKS**

Numbers from 1 to 50 are written on slips of paper and put in a box. Without looking, a slip is to be drawn from it.

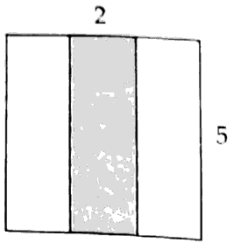
(a) What is the probability that it is a multiple of 4 ?

(b) What is the probability that it is a multiple of 6 ?

(c) What is the probability that it is a multiple of 4 and 6 ?

6. 2024 **2 MARKS**

A rectangular portion is shaded in a square of side 5 centimetres as shown in the figure. A dot is put inside the square without looking. Find the probability of the dot to be in the shaded region.



7. 2023 **4 MARKS**

A box contains 4 slips numbered 1, 2, 3, 4 and another contains 5 slips numbered 1, 2, 3, 4, 5. One slip is taken from each box without looking it.

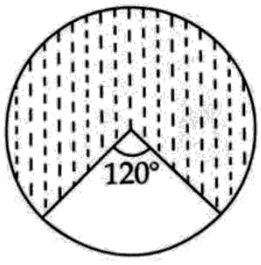
(a) In how many different ways we can choose the slips ?

(b) What is the probability of both numbers being odd ?

(c) What is the probability of both numbers being the same ?

8. 2023 **2 MARKS**

A dot is put inside the circle without looking it.



(a) What is the probability that the dot to be within the unshaded part ?

(b) What is the probability that the dot to be within the shaded part ?

9. 2022 **4 MARKS**

A box contains four slips numbered 1, 2, 3, 4 and another box contains five slips numbered 5, 6, 7, 8, 9. If one slip is taken from each box.

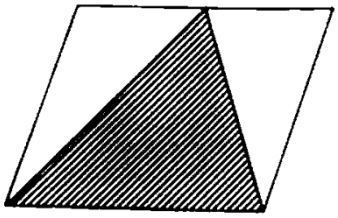
(a) How many number pairs are possible ? **(1)**

(b) What is the probability of both being odd ? **(1)**

(c) What is the probability of getting the sum of the numbers 10 ? **(2)**

10. 2022 **2 MARKS**

A triangle is drawn by joining the mid-point of one side of a parallelogram and the endpoints of the opposite side. The triangle is shaded as shown in the figure.



(a) What is the area of the triangle, if the area of the parallelogram is 50 square centimetres ? **(1)**

(b) Find the probability of a dot put without looking, to be within the triangle. **(1)**

11. 2022 **1 MARK**

A box contains 7 white balls and 3 black balls. If a ball is taken from it, what is the probability of it being black ?

12. 2021 **4 MARKS**

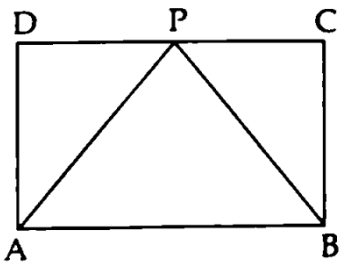
One is asked to say a two-digit number,

(a) What is the probability of both digits being the same ? **(2)**

(b) What is the probability of the first digit being twice the second ? **(2)**

13. 2021 **3 MARKS**

ABCD is a rectangle. P is the mid-point of CD. If we put a dot in the figure without looking into it :



(a) What is the probability that it would be inside triangle APB ? **(1)**

(b) What is the probability that it would be inside triangle ADP ? **(2)**